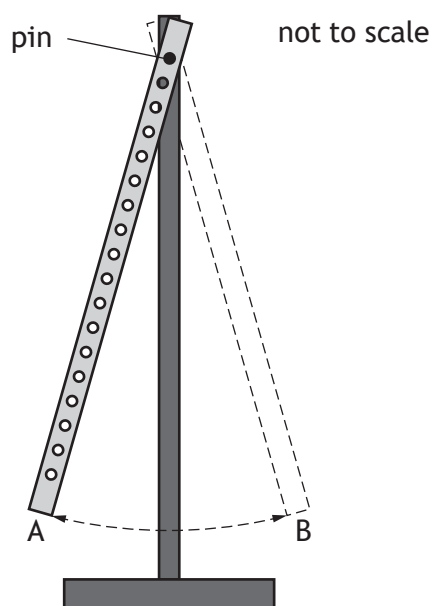


15. A 1.00 m long wooden rod has a series of small holes drilled at 10 mm intervals along its length. The rod is hung on a horizontal pin passing through a hole 50 mm from one end.



The rod is then raised through a small angle and released.

The period T is the time for the rod to travel from A to B and back to A.

- (a) Describe a method to obtain an accurate value for the period T using only a stopwatch.

2



15. (continued)

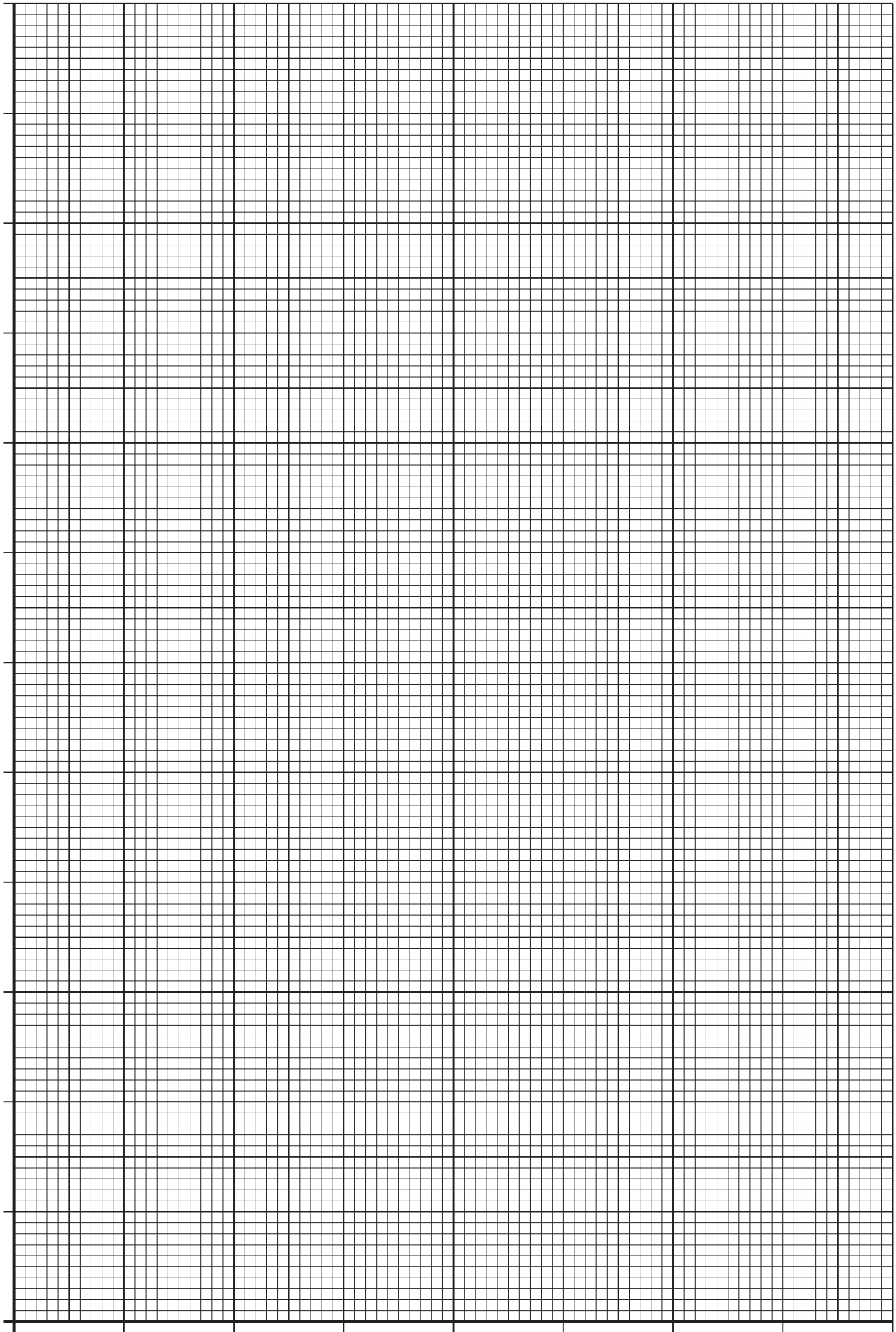
- (b) The rod is hung from different holes in turn, and the distance h from the pin to the midpoint of the rod is recorded.

T is determined for each value of h . The results are shown in the table.

h (m)	T (s)
0.45	1.60
0.40	1.56
0.35	1.54
0.30	1.53
0.25	1.53
0.22	1.55
0.20	1.58

- (i) Using the square-ruled paper on *page 41*, draw a graph of T against h . 3
- (ii) Using your graph, state the **two** values of h that produce a period of 1.57 s. 1
- (iii) (A) Using your graph, estimate the minimum period T . 1
- (B) Suggest an improvement to the experimental procedure that would allow a more precise value for the minimum period T to be determined. 1





* X 8 5 7 7 6 0 1 4 1 *

15. (continued)

(c) The quantities T and h are related by the relationship

$$T^2 h = \frac{4\pi^2 h^2}{g} + C$$

where g is the gravitational field strength and C is a constant.Use data from the table on **page 40** to calculate a value for C when h is 0.30 m.

A unit is not required.

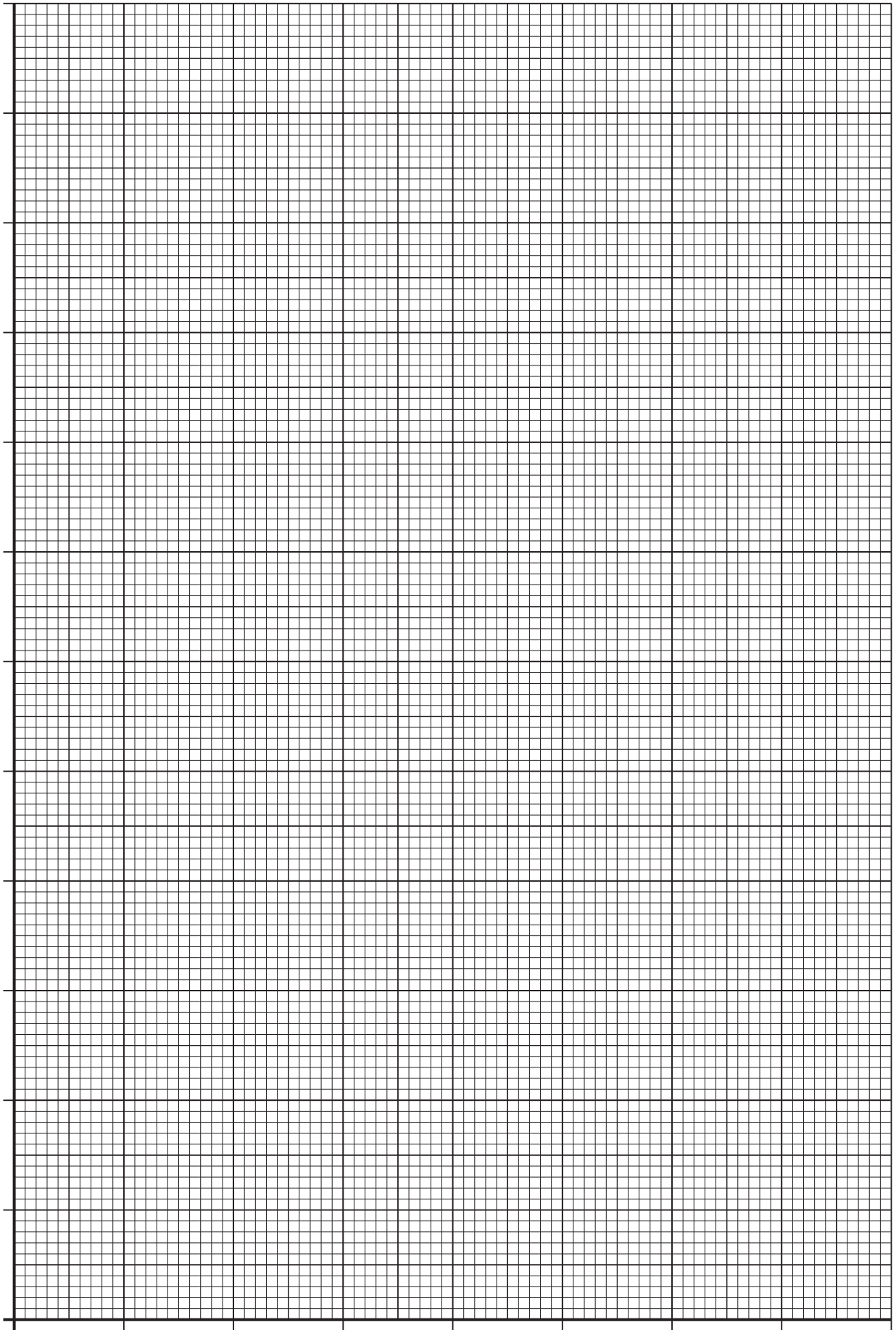
2

Space for working and answer

[END OF QUESTION PAPER]



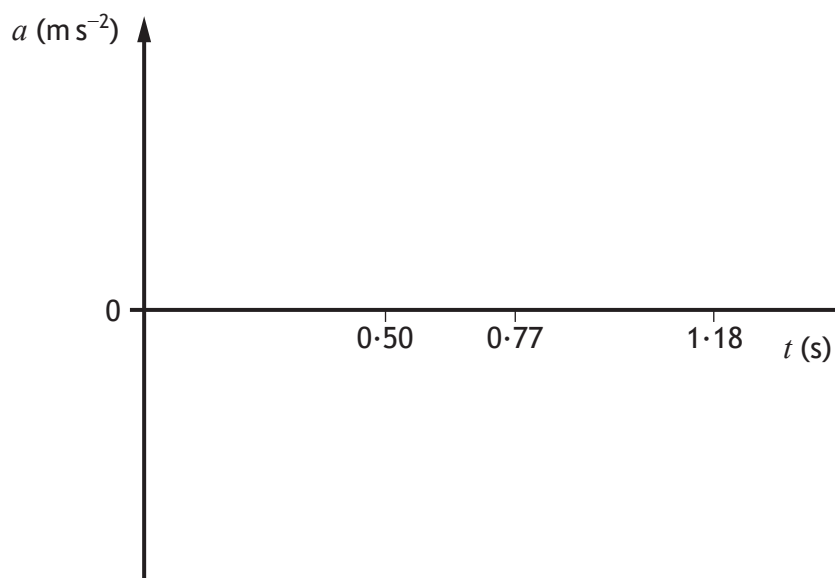
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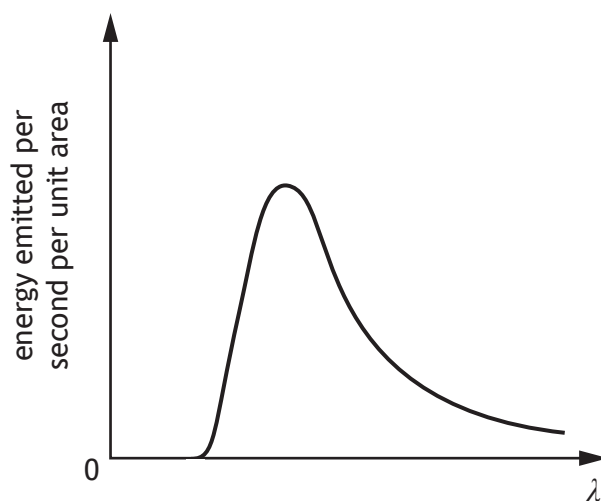
* X 8 5 7 7 6 0 1 4 3 *

ADDITIONAL SPACE FOR ANSWERS AND ROUGH WORK

Additional graph for use with Question 1 (c)



Additional graph for use with Question 6 (a)



* X 8 5 7 7 6 0 1 4 4 *

ADDITIONAL SPACE FOR ANSWERS AND ROUGH WORK

Additional graph for use with Question 13 (b)

